

Calling the Table

A Poisson–Monte Carlo Model of the Premier League

Group: [MEMBER 1] — Group Lead; [MEMBER 2] — Project Planner; **Ankit Karki — QA Lead.**

Prepared: October 2, 2026. **Instructor:** [INSTRUCTOR]. **Canvas deadline:** [fill in].

1 Deliverables - What You Submit

This report follows Results Requirements.pdf in the repository. It uses the proposal as its scope baseline and the delivered files as evidence. The proposal PDF is still labeled **group-review draft**; instructor approval evidence is [not supplied]. “Yes” means an artifact exists, not that a student signed it. “Partial” and “No” retain the explanation required by the rubric.

Promised	Delivered	Evidence file or section	Explanation if Partial or No
Three members and assigned roles	Partial	§1; proposal §1	Ankit is QA Lead; [MEMBER 1] (Lead), [MEMBER 2] (Planner) still need names/confirmation.
Signed group contract and personal peer forms	No	Proposal submission checklist	No completed signed artifacts verified; outside this report-writing request.
Topic uniqueness and instructor proposal sign-off	No	Proposal submission checklist	No verification supplied; proposal PDF itself is marked group-review draft.
Milestone plan with proposed and actual dates	Partial	§2	Proposed windows reproduced; each Actual date and major-deviations account awaits group input.
Poisson/core-model/points/Monte Carlo explanation	Yes	§3; docs/Mathematical_Notes.md	
Five worked numerical examples and independent checks	Partial	§3; outputs/hand_checks.csv; qa/Ankit_Karki_QA_Workbook.xlsx	Inputs/cutoffs and formula calculations exist; personal rate/probability review is unsigned.
Data dictionary, provenance and validation findings	Yes	§3; data/provenance.json; outputs/validation.json	
Fitted team rates and strengths	Yes	outputs/fitted_strengths_halfway.csv	
Selected scoreline matrix and analytic/simulation comparison	Partial	§3; outputs/selected_score_matrix.csv; qa_checks.json	Numerical comparison delivered; PNG extent label needs correction (QA-017).
Walk-forward predictions and proper-score tables	Yes	§3; outputs/walk_forward_predictions.csv; walk_forward_scores.csv	
Paired bootstrap intervals and calibration plots	Yes	§3; outputs/bootstrap_m0_minus_m1.json; calibration_m1.csv	
Halfway table, points intervals and actual comparison	Yes	§3; outputs/halfway_forecast_table.csv	
Success/failure/limitation discussion	Yes	§3; docs/Results_Report.md	
Reproducible program source and setup instructions	Yes	§4; src/standings/; docs/Getting_Started.md	
Python/pandas/NumPy/SciPy/matplotlib/GitHub tools	Yes	§4; outputs/source_metadata.json; repository	
Spreadsheet checks, milestone tracking and QA review	Partial	§2; qa/Ankit_Karki_QA_Workbook.xlsx	Calculated workbook exists; actual milestone dates and human review remain open.
VS Code or optional Colab workflow	Partial	docs/Getting_Started.md	Instructions exist; actual use/version [not recorded]; optional environment, not a core result.

Reconciliation source: proposal §§1–5 and submission checklist; evidence read from the existing repository and output files. No personal Peer Evaluation, Group Contract or Individual Reflection is written here.

Deliverables – reconciliation continued

Promised	Delivered	Evidence file or section	Explanation if Partial or No
PowerPoint-format class-presentation draft	Yes	§5; docs/Presentation_Resource.md	Nine-slide editable Manus PPTX-mode draft; no standalone .pptx file was verified.
Scheduled class presentation and individual contributions	No	§5; proposal §5	Instructor scheduling, rehearsal and delivered contributions not verified.
QA plan, test evidence and dated issue log	Yes	§6; qa/Test_Plan.md; outputs/test-results.txt; qa/Issue_Log.csv	
Data/goal/result/fixture checks	Yes	§6 D01–D05; outputs/validation.json	
Complete independent official-table reconciliation	Partial	§6 D06; qa/Independent_Table_Comparison.csv	NBC: 119/120 fields agree; Fulham +4 vs –4 retained; GF/GA and official checks incomplete.
Probability, leakage and simulation checks	Partial	§6 M03/P01/S01–S06; qa_checks.json	Core evidence delivered; some planned edge tests and personal review not verified.
Stability, scoring, calibration and goal-count/coverage diagnostics	Yes	§3; §6 E01–E06	Reported honestly; negative predictive results are not forced passes. Planned score edge tests remain partial in E01.
Consistent documents/figures/presentation and QA sign-off	Partial	§6 R02–R03	Known stale text and matrix-label issue disclosed; personal signature/approval left blank.
Requirement checklist and Part 2 reconciliation	Partial	§1; §6; docs/Requirements_Checklist.md	This report reconciles Part 2; older checklist/draft missing-document statements are stale and retained.
2026/27 genuinely prospective forecast log/test (optional)	No	§3; QA-010	50 historical results are not a prospective test; no real saved future log verified.
Fitted Dixon–Coles Model 2 (optional)	No	Proposal Model 2; docs/Results_Report.md	Stretch goal not implemented; core retained without it.
Shrinkage tuning/comparison (optional experiment)	No	source_metadata.json	Run shrinkage=0; no tuning experiment or independent evaluation delivered.

2 The High-Level Project Plan

The proposed windows below are copied from the proposal, not asserted as course deadlines. File/run timestamps are not a substitute for a member-confirmed completion record. Every Actual date is intentionally left for the group.

Proposed window	Milestone / evidence	Actual date
Oct 2–9	Topic, roles, contract, due date, proposal review/sign-off, obtain Results Requirements	[fill in]
Oct 10–23	Freeze/validate data; Models 0/1, grids, five hand checks	[fill in]
Oct 24–Nov 6	Simulations, reproducibility, walk-forward and scored comparisons	[fill in]
Nov 7–13	Table backtest, calibration, first results and presentation drafts	[fill in]
Nov 14–20	Error review, repairs, table/graph/presentation verification	[fill in]
Nov 21–Dec 1	Thanksgiving buffer; optional extensions; final QA/rehearsal	[fill in]
Dec 2, 2026 (proposed)	Class presentation and results submission, subject to instructor confirmation	[fill in]

Major deviations from the plan: [fill in]. Record what changed, why, its effect, and any instructor-agreed change in scope. Confirm calendar constraints and the actual presentation/submission date with the instructor; the proposed December 2 date is not approval.

3 Written Documentation

3.1 Data, chronology and model

The completed 2025/26 snapshot contains **380 matches and 20 teams**, from August 15, 2025 through May 24, 2026. Complete-season validation records each directed pairing once and 19 home/19 away matches per team. Whole-season goal means, 1.5263 home and 1.2237 away, are descriptive only. Forecasts use strictly earlier dates. After a 100-match warm-up, the stored evaluation has **280** valid common-odds rows and no excluded odds rows.

Field	Meaning and use
Date	Match date; training uses Date strictly before prediction Date.
HomeTeam / AwayTeam	Team identities; home/away orientation is preserved.
FTHG / FTAG	Full-time home/away integer goal counts.
FTR	Observed H, D or A result, checked against the goals.
AvgCH / AvgCD / AvgCA	Fixed market-average closing decimal odds; used only for the reference.

The data come from football-data.co.uk [3]. Pinnacle columns are excluded under the provider's stale-API warning. The 2026/27 snapshot has 50 past results through September 20, 2026; it is **not** a prospective test. Its xG fields are deliberately unused. `outputs/fitted_strengths_halfway.csv` contains venue-specific rates and strength ratios at the frozen boundary.

For home goals X and away goals Y , the independent-Poisson approximation is [1,4]

$$X \sim \text{Poisson}(\lambda_H), \quad Y \sim \text{Poisson}(\lambda_A), \quad P(X = k) = e^{-\lambda_H} \frac{\lambda_H^k}{k!}.$$

Model 0 sets $\lambda_H = \mu_H$ and $\lambda_A = \mu_A$ from the training league means. For Model 1, s denotes scoring and c conceding rates, using the indicated venue:

$$\lambda_H = \frac{s_{H(h)} c_{A(a)}}{\mu_H}, \quad \lambda_A = \frac{s_{A(a)} c_{H(h)}}{\mu_A}.$$

This is an unsmoothed method-of-moments ratio model, not Maher's full regression. Missing histories use the corresponding training league rate. Zero-rate cases are handled explicitly. Independence and frozen strength are assumptions, not established facts.

Fully worked fixture — Tottenham–Man United, November 8, 2025. The 100-match training set ends November 3. League sums are 155 home goals and 113 away goals. Tottenham's home scoring and conceding sums are 5 and 5 over 5 fixtures; Man United's away scoring and conceding sums are 6 and 10 over 5 fixtures. Thus

$$\lambda_H = \frac{\left(\frac{5}{5}\right)\left(\frac{10}{5}\right)}{\frac{155}{100}} = 1.290323, \quad \lambda_A = \frac{\left(\frac{6}{5}\right)\left(\frac{5}{5}\right)}{\frac{113}{100}} = 1.061947.$$

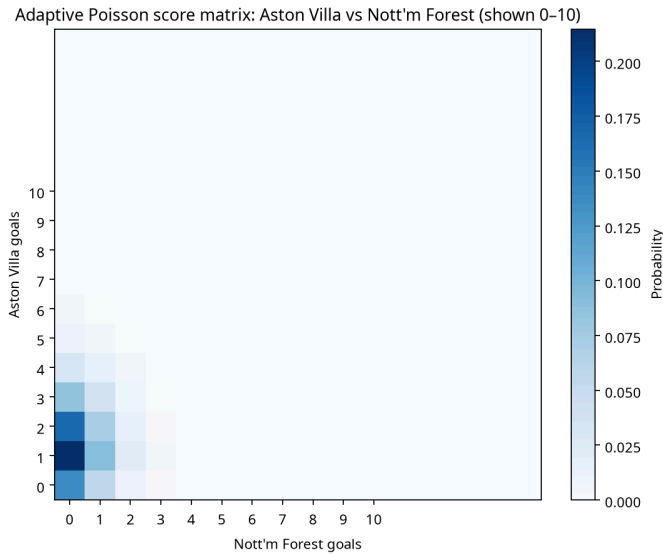
Aggregating the joint score distribution gives stored Model 1 H/D/A probabilities **0.416932 / 0.277617 / 0.305450**. A predicted mean is not a certain score. The five training-safe checks are below; the fractions show the exported sums and counts rather than rounded inputs.

Date / fixture	Latest training / N	Home expected-goals calculation	Away expected-goals calculation
2025-11-08 Tottenham–Man United	2025-11-03 N=100	$(5/5) \times (10/5) \div (155/100) = 1.290323$	$(6/5) \times (5/5) \div (113/100) = 1.061947$
2025-12-22 Fulham–Nott'm Forest	2025-12-21 N=169	$(14/8) \times (12/8) \div (272/169) = 1.630974$	$(6/8) \times (10/8) \div (208/169) = 0.761719$
2026-02-02 Sunderland–Burnley	2026-02-01 N=239	$(18/11) \times (27/11) \div (372/239) = 2.580512$	$(13/11) \times (9/11) \div (294/239) = 0.786052$
2026-04-10 West Ham–Wolves	2026-03-22 N=309	$(18/15) \times (23/15) \div (463/309) = 1.227991$	$(7/15) \times (28/15) \div (382/309) = 0.704642$
2026-05-24 West Ham–Leeds	2026-05-19 N=370	$(24/18) \times (32/18) \div (567/370) = 1.546803$	$(20/18) \times (30/18) \div (454/370) = 1.509218$

Sources: `validation.json`, `walk_forward_metadata.json`, `hand_checks.csv` and `walk_forward_predictions.csv` in `outputs/`. The workbook's calculated rates agree; Ankit's personal input/probability review remains pending.

Written Documentation — scoreline matrix and simulation stability

The selected matrix fixture is **Aston Villa–Nott'm Forest, January 3, 2026**, using the frozen January 1 fit: $\lambda_H = 1.5452952953$, $\lambda_A = 0.4283413849$. It is different from the first worked fixture. The normalized joint cell probability is $g_{ij} = P(X = i)P(Y = j)$. Home win sums $i > j$ cells, draw sums $i = j$, and away win sums $i < j$. The grid begins at 0–10 and expands; the saved matrix spans 0–15 in both directions.



Numeric low-score corner. Rows are home goals; columns are away goals. Values below are cells of the full normalized matrix, not a renormalized 0–4 submatrix.

H / A	0	1	2	3	4
0	0.138951	0.059518	0.012747	0.001820	0.000195
1	0.214720	0.091973	0.019698	0.002812	0.000301
2	0.165903	0.071063	0.015220	0.002173	0.000233
3	0.085456	0.036604	0.007840	0.001119	0.000120
4	0.033014	0.014141	0.003029	0.000432	0.000046

Source: outputs/selected_score_matrix.csv.

Figure 1 shows that probability is concentrated in low scores, but the model retains rare higher scores for accurate aggregation. The figure label will be fixed later by cropping to 0–10 or relabeling the full extent, not silently changing this frozen evidence.

Figure 1: Saved selected-fixture heatmap, retained unchanged. Its “shown 0–10” title is misleading: the CSV and displayed adaptive matrix span 0–15 (QA-017). Source: outputs/figures/selected_score_matrix.png.

Method	P(home win)	P(draw)	P(away win)
Adaptive grid	0.652989336	0.247310473	0.099700190
Skellam	0.652989336	0.247310473	0.099700190
Simulation, N=100,000	0.65121	0.24913	0.09966

Skellam checks the goal difference $X - Y$ [4]. The selected grid/Skellam maximum discrepancy is 4.111×10^{-12} . Simulation is stochastic and is not expected to match every digit. Across the recorded run, maximum omitted grid mass was 9.668×10^{-9} , below the 10^{-8} tolerance.

For an outcome frequency $\hat{p}$ in N simulations, the diagnostic uses

$$SE = \sqrt{\frac{p(1-p)}{N}}, \quad z = \frac{\hat{p} - p}{SE},$$

where p is the analytic probability. The file stores the **maximum absolute** standardized discrepancy across H/D/A for each run size:

Runs N	Home freq.	Draw freq.	Away freq.	Max z	All ≤ 3 SE?
1,000	0.66200	0.23100	0.10700	1.195466	Yes
5,000	0.64740	0.24320	0.10940	2.289321	Yes
10,000	0.65590	0.24650	0.09760	0.700999	Yes
100,000	0.65121	0.24913	0.09966	1.333611	Yes

All four saved flags are favorable, but they are not a promise for every seed. The standardized discrepancy is not monotonically smaller as N grows. Seed 3315 is recorded; more samples reduce Monte Carlo error, not model error. In the season calculation, a win contributes 3 points, a draw 1 each, and a loss 0; expected home points are $3p_H + p_D$.

Sources: outputs/qa_checks.json and selected_score_matrix.csv; SciPy probability references [4]. No model or simulation was rerun for this report.

Written Documentation — walk-forward scores and paired uncertainty

On each evaluation date, all forecasts use only results on strictly earlier dates. A complete warm-up date batch is excluded, preventing same-date leakage. All three methods are compared on the same **280** valid AvgC rows. Model/market probabilities are in H/D/A order. For observed one-hot outcome y , the implemented scores are

$$\text{Brier} = \sum_{c=1}^3 (p_c - y_c)^2, \quad \text{RPS} = \frac{1}{2} \sum_{j=1}^2 \left(\sum_{c=1}^j p_c - \sum_{c=1}^j y_c \right)^2.$$

Brier is unscaled; RPS is divided by two. Lower values are better. Market probabilities are $p_c^{\text{market}} = \frac{\frac{1}{o_c}}{\sum_j \frac{1}{o_j}}$, with complete finite odds greater than one.

Model	N	Mean Brier	Mean RPS
M0	280	0.667304	0.228289
M1	280	0.661714	0.222924
Market	280	0.620037	0.204016

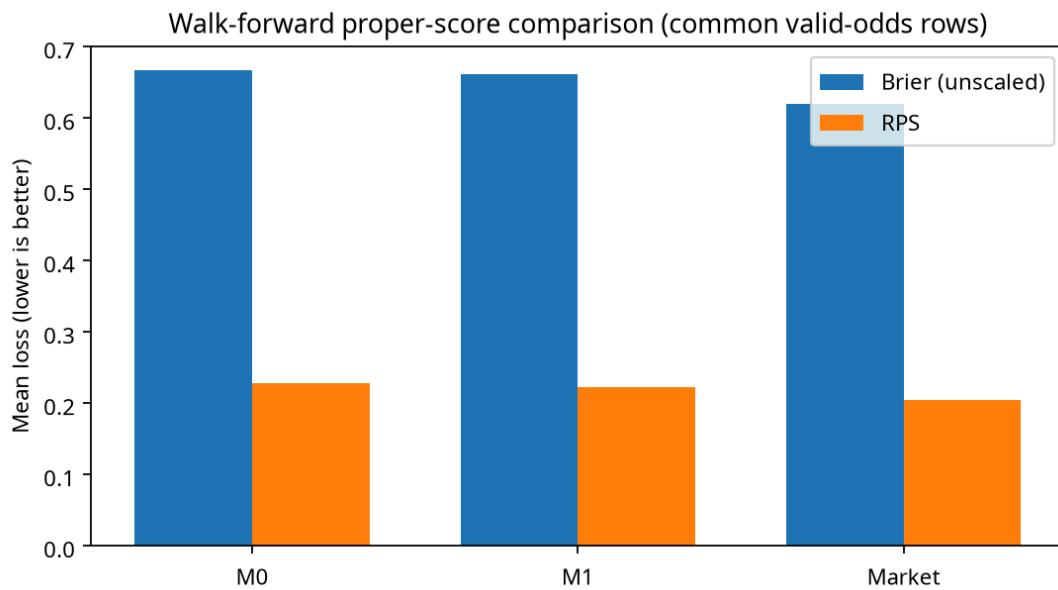


Figure 2: Stored common-sample proper-score comparison. Brier and RPS have different scales; compare methods within a metric. Source: [outputs/figures/score_comparison.png](#).

Figure 2 shows a small M1 improvement in point estimates and lower market losses. The date-block bootstrap preserves paired forecasts and same-date groups. The contrast is **M0 minus M1**, so positive values favor M1. There are 85 date blocks, with 2,000 replicates per metric.

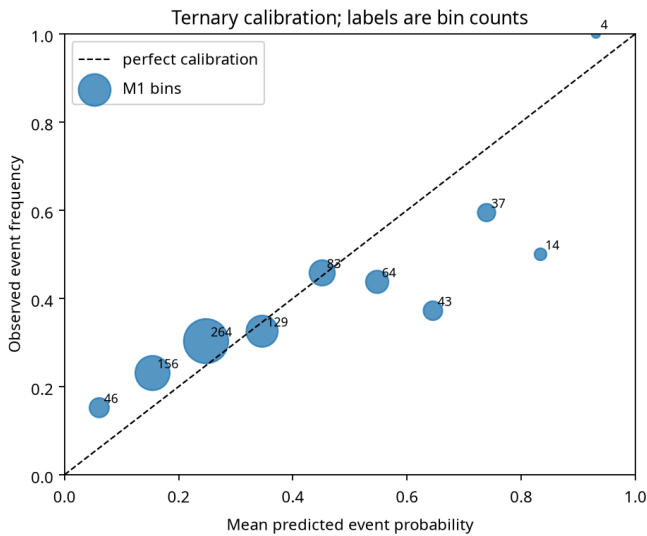
Metric	Mean M0–M1	95% paired interval	Replicates	Seed
BRIER	+0.005590	[-0.031277, +0.042015]	2000	3315
RPS	+0.005366	[-0.010215, +0.021621]	2000	3316

Model 1's improvement over Model 0 is not established: both intervals include zero. This is not proof of equal performance. The market benchmark scored best, but closing odds contain information nearer kickoff than a historical goals-only model. It is a contextual reference, not an equal-information experiment, and inverse-odds normalization need not yield unbiased probabilities. No betting conclusion is drawn.

Sources: [outputs/walk_forward_scores.csv](#), [walk_forward_row_losses.csv](#), [walk_forward_metadata.json](#) and [bootstrap_m0_minus_m1.json](#).

Written Documentation — calibration and goal-count diagnostic

Calibration compares forecasted probabilities with observed event frequencies. Figure 3 pools the home/draw/away event predictions; the bin count table is taken from the underlying CSV. A match contributes three related events, so the 840 event predictions are **not** 840 independent matches. Sparse high-probability bins should not support strong conclusions.



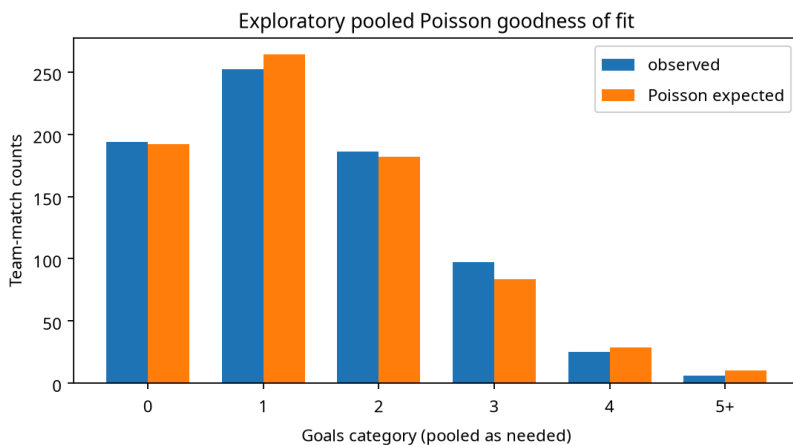
Probability bin	Count	Mean p	Observed
0.0–0.1	46	0.0609	0.1522
0.1–0.2	156	0.1542	0.2308
0.2–0.3	264	0.2480	0.3030
0.3–0.4	129	0.3461	0.3256
0.4–0.5	83	0.4514	0.4578
0.5–0.6	64	0.5477	0.4375
0.6–0.7	43	0.6453	0.3721
0.7–0.8	37	0.7393	0.5946
0.8–0.9	14	0.8338	0.5000
0.9–1.0	4	0.9308	1.0000

Source: outputs/calibration_m1.csv.

Several upper bins have observed frequencies below predicted means, suggesting possible overconfidence in this sample. The 0.9–1.0 bin has only four events; its observed frequency of 1.0 is not persuasive evidence of precise calibration.

Figure 3: Stored pooled event calibration. Labels are bin counts; dashed line is perfect calibration. Source: outputs/figures/calibration_m1.png.

The exploratory goal-count diagnostic pools 760 team-match goal counts, estimates a single league-wide rate $\lambda = 1.375$, and compares observed counts with Poisson expectations. Figure 4 and the table preserve the pooled bins; all expected counts are at least five.



Goals	Observed	Expected
0	194	192.158
1	252	264.217
2	186	181.649
3	97	83.256
4	25	28.619
5+	6	10.100

Figure 4: Observed versus expected goal counts for the pooled diagnostic. Source: outputs/figures/goals_goodness_of_fit.png.

The saved Pearson statistic is **5.077608** and the refitted-lambda parametric-bootstrap p-value is **0.2837**, from 1,000 replicates, seed 3315. Each reference sample refits lambda. The result does not show strong disagreement for this pooled diagnostic, but it does **not** establish independent identical team scoring or out-of-sample predictive accuracy. Team heterogeneity and within-match dependence remain model limitations.

Sources: outputs/goals_goodness_of_fit.csv and .json; calibration_m1.csv. Statistical reference simulations are not substitute research datasets.

Written Documentation — halfway season forecast and interval misses

The fit was frozen at the first complete date at/after 190 matches: **January 1, 2026**, with **190 played and 190 remaining**. The 10,000-season simulation (seed 3315, shrinkage zero) uses future pairings retrospectively but never their actual future goals in fitting. Final-points **MAE = 5.358**; **16 of 20** actual totals fall within the nominal 90% intervals (**80%** descriptive coverage).

Team	Expected points	90% interval	Actual points	Inside?
Arsenal	85.884	78–93	85	Yes
Man City	82.760	75–91	78	Yes
Aston Villa	72.022	63–81	65	Yes
Chelsea	61.947	53–71	52	No
Liverpool	59.982	51–69	60	Yes
Man United	57.664	49–67	71	No
Newcastle	54.679	46–63	49	Yes
Tottenham	54.085	45–63	41	No
Everton	54.349	46–63	49	Yes
Sunderland	53.978	46–62	54	Yes
Crystal Palace	52.835	44–62	45	Yes
Brentford	52.346	44–60	53	Yes
Fulham	52.151	44–61	52	Yes
Brighton	50.599	42–59	53	Yes
Bournemouth	50.462	42–59	57	Yes
Leeds	45.914	38–54	47	Yes
Nott'm Forest	35.041	27–44	44	Yes
West Ham	33.138	25–42	39	Yes
Burnley	27.761	20–36	22	Yes
Wolves	12.189	6–19	20	No

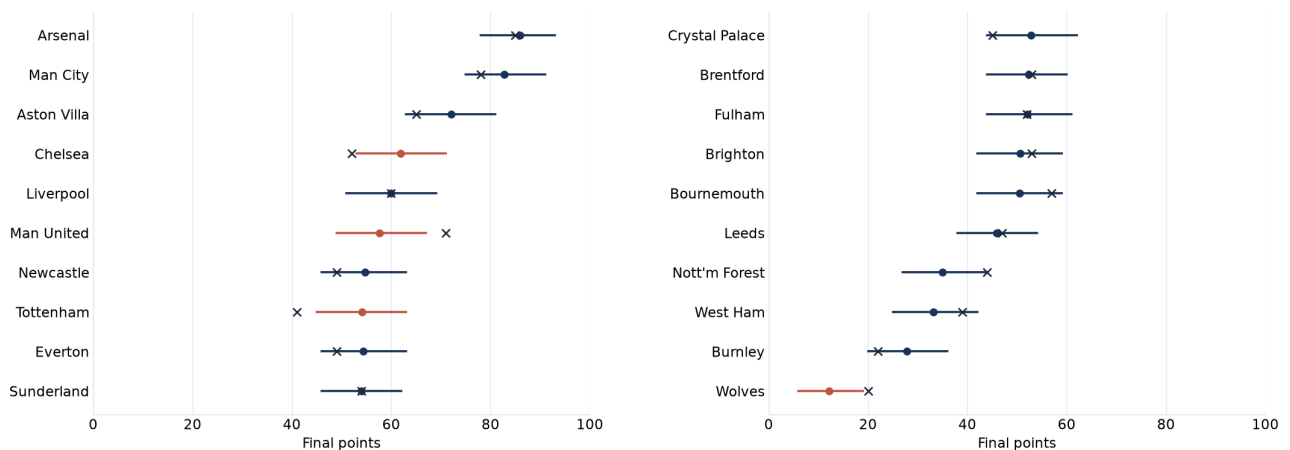


Figure 5: Stored two-panel all-team comparison also used by the draft slides. Dots: simulated means; lines: 90% intervals; crosses: actual totals. Red identifies the four interval misses. Source: `outputs/figures/presentation_forecast_intervals.png`.

Figure 5 retains the failures. **Chelsea** (52 versus 53–71) and **Tottenham** (41 versus 45–63) finished below their intervals; **Man United** (71 versus 49–67) and **Wolves** (20 versus 6–19) finished above. Static strength and omitted information can contribute to misses, but no cause is established for an individual club. Intervals condition on fitted rates and omit parameter/structural uncertainty. Twenty correlated team totals are not independent coverage trials. Rows follow expected simulated rank, not descending mean points. Rankings use points/GD/GF; unresolved ties get average ranks and fractional boundary weights, not official head-to-head decisions. Top-four finish probability is not a universal qualification guarantee.

Source: `outputs/halfway_forecast_table.csv` and `qa_checks.json`.

4 Computational/Technology Tools

The recorded environment below comes from `outputs/source_metadata.json`, not a guessed current version. The source run used seed 3315, 10,000 seasons, zero shrinkage, warm-up 100 and halfway target 190. The recorded UTC run timestamp is October 2, 2026, 22:38:05; it does not fill an actual milestone date.

Tool	Recorded version
python	3.12.3
numpy	2.5.3
pandas	3.0.6
scipy	1.18.1
matplotlib	3.11.2
tabulate	0.10.0

Python orchestrates the pipeline; pandas validates/filters results, NumPy samples and accumulates, SciPy supplies distributions/diagnostics, matplotlib creates plots, and tabulate formats tables. The spreadsheet's software version is [not recorded]. GitHub stores source/version history and the test workflow. Optional editor/Colab use and version are [not recorded].

Repository layout. `data/` contains provenance/private snapshots; `src/standings/` contains the program; `tests/` contains synthetic-only unit inputs; `outputs/` contains frozen generated evidence; `qa/` contains the plan/log/workbook; `docs/` contains proposal, notes and drafts; `.github/workflows/tests.yml` configures unit tests. Source files in `src/standings/` are:

Source file	One-line responsibility
<code>__init__.py</code>	Package identity and public model helpers.
<code>__main__.py</code>	Entry point for <code>python -m standings</code> .
<code>cli.py</code>	Commands, pipeline orchestration, table exports and run metadata.
<code>data.py</code>	Strict result validation and deterministic table reconstruction.
<code>models.py</code>	Training-only league and venue-specific scoring/conceding rates.
<code>probability.py</code>	Adaptive joint score grid, H/D/A sums and Skellam check.
<code>evaluation.py</code>	Date-safe predictions, fixed odds, proper scores and paired bootstrap.
<code>simulation.py</code>	Frozen remaining fixtures, points/tie weights and stochastic QA.
<code>diagnostics.py</code>	Calibration bins and refitted-rate pooled Poisson diagnostic.
<code>reporting.py</code>	CSV/JSON/Markdown report formatting and plot generation.
<code>live.py</code>	Manual download, immutable logs and chronology-validated scoring.

Reproduce without overwriting delivered outputs. From the `results-report` branch, with Python 3.11+ and a local historical CSV:

```
python -m pip install -e '.[dev]'
python -m pytest -q
python -m standings run --data data/raw/E0_2526.csv \
  --out outputs_repeat --seed 3315 --simulations 10000
```

The tested direct versions are in `requirements-lock.txt`. Any repeat must be compared with the delivered files; it must not overwrite `outputs/`. This report only audited those files; it did not rerun `pytest` or the model.

The older `outputs/analysis_2526/` copies were also inspected. Their common numerical CSV content agrees with the active files, but some metadata and hand-check fields are less complete. This report uses the top-level `outputs/` evidence and does not combine old/new provenance descriptions.

Historical input SHA-256: `3e3a8352f9ada6789c508d6ca184424421fed56a30400904a4a327c583407e62` The current local hash matches the source record. `data/provenance.json` records acquisition; a hash identifies bytes, not every historical remote retrieval route. Raw provider CSVs remain private coursework copies and are ignored in public Git. Attribute `football-data.co.uk` and follow its conditions [3]; do not republish raw data or turn the one-off manual download into an unattended scraper.

5 Presentation

The existing **nine-slide editable PPTX-mode draft** is identified in docs/Presentation_Resource.md. Its authored source/outline was inspected; no standalone exported .pptx file or actual PowerPoint playback was verified. It remains a draft, not an already-delivered class talk.

Slide	Title	Content / existing figure
1	Calling the Table	Title, course, Ankit's QA role and draft status.
2	One question, one completed season	Question, 380 matches/20 teams and descriptive means.
3	From goals to probabilities	Core formulas and the Tottenham–Man United example.
4	Time is the train/test boundary	Warm-up, strict dates, halfway boundary and seeds.
5	A small gain—not conclusive evidence	Native score tables and bootstrap intervals; no PNG.
6	The table is a distribution	presentation_forecast_intervals.png; all 20 clubs.
7	QA checks implementation—not wishful outcomes	calibration_m1.png plus recorded QA metrics.
8	Where this model stops	Assumptions, omitted factors and human tasks.
9	Reproduce it and verify it	Commands, references and preserved external discrepancy.

The only PNGs embedded in the draft are the two-panel points figure on slide 6 and the calibration figure on slide 7. The score slide uses editable native tables. Other numerical/model text follows the existing output evidence. The group must still add its real names, rehearse and verify the final exported deck. **Class presentation time must still be scheduled with the instructor.** Presentation time: [fill in]. Duration: [fill in]. Instructor-review appointment: [fill in]. The rubric encourages a review of the results and presentation before speaking to the class.

Existing draft: [Calling the Table — Premier League Modeling Project](#). No deck content was altered.

6 Quality Assurance Signoff

6.1 Test Results

outputs/test-results.txt records **38 pytest tests passed in 7.98 seconds**. Parametrized cases explain why there are fewer named test functions than passed cases. Passing that suite does not establish every manual or planned edge check. The table audits **all 31** cases in qa/Test_Plan.md: PASS means the stated evidence, DIAGNOSTIC is statistical evidence, and PARTIAL/PENDING is not personal sign-off.

QA case / check	Status	Observed evidence / remaining gap
D01 Required schema	PASS (recorded)	validation.json: required fields and valid dates; loader/test evidence.
D02 Nonnegative integer goals	PASS (recorded)	Strict loader; decimal-goal rejection test; complete results validated.
D03 Goals agree with FTR	PASS (recorded)	validation.json complete run; inconsistent/invalid FTR tests.
D04 Duplicate/self fixtures	PASS (data)	Complete run and duplicate test; no duplicate/self fixtures in retained season.
D05 Complete season	PASS (recorded)	380/20; each directed pair once, 19 home/19 away; complete-schedule test.
D06 Official table	PARTIAL / PENDING	119/120 NBC fields agree; Fulham sign differs; official GF/GA/P and adjustments not verified.
D07 Provenance	PASS (hash/policy)	Local hash matches source_metadata.json; provenance record and ignored raw CSVs present.
M01 League means	PASS (recorded)	Training-only means in prediction rows; full-season means descriptive only.
M02 Five independent checks	PARTIAL	Workbook cached formulas have no errors and rate differences <0.0005; personal probability/input review unsigned.
M03 Missing/zero histories	PARTIAL	Zero-rate tests pass; fallback/empty-training logic exists, but dedicated unseen-team/empty-training tests not verified.

Quality Assurance Signoff — Test Results continued

QA case / check	Status	Observed evidence / remaining gap
P01 Orientation/symmetry	PARTIAL	M1 orientation test and lower-triangle aggregation; dedicated equal-rate symmetry test not verified.
P02 Mass and tolerance	PASS (recorded)	Finite valid probabilities; sums within 1e-12; maximum tail 9.668×10^{-9} ; adaptive tests.
P03 Analytic cross-check	PASS (recorded)	Grid/Skellam selected discrepancy 4.111×10^{-12} ; single/double-zero tests.
L01 Strict prior dates	PASS (recorded)	Every stored latest_training_date < Date; warm-up batch and later-result invariance tests.
S01 Match Monte Carlo	DIAGNOSTIC	All four stored max z values <3; see stability table. A favorable random result is not a universal pass.
S02 Reproducibility	PASS (recorded)	Seeded match/season tests; root/archive forecast CSVs byte-identical. Full simulation arrays not saved.
S03 Simulation stability	DIAGNOSTIC	N=1,000/5,000/10,000/100,000 stored; nonmonotonic standardized deviations disclosed.
S04 3/1/0 points	PASS (recorded)	qa_checks.json conservation true and simulation conservation test.
S05 Season cutoff	PASS (recorded)	Jan 1, 2026: 190 played/190 remaining; full-date boundary test; fixed rates.
S06 Ranks/ties/weights	PASS (model rule)	Saved title/top-four/relegation weights sum to 1/4/3; average/fractional ties are an approximation.
E01 Brier/RPS convention	PARTIAL	Saved definition tests verify a nontrivial H outcome; planned perfect/uniform edge tests not explicitly verified.
E02 Fixed odds/common sample	PASS (recorded)	AvgCH/AvgCD/AvgCA; 280 valid common rows, 0 exclusions; invalid-triplet test.
E03 Paired date bootstrap	PASS (recorded)	2,000 replicates, 85 date blocks; seeds 3315/3316; paired contrast and reproducibility test.
E04 Calibration	PASS (data/visual)	Bin counts/frequencies match plot; related event predictions and sparse bins identified.
E05 Goal-count diagnostic	DIAGNOSTIC	Pooled expected counts ≥ 5 ; $\chi=1.375$; refitted bootstrap $p=0.2837$; not a proof of independence.
E06 Points coverage	DIAGNOSTIC	16/20 (80%) inside nominal 90% intervals; MAE 5.358; all four misses retained.
V01 Genuine live chronology	PARTIAL (tests only)	Synthetic log tests include real-creation cutoff and spoofed-status recomputation; no real prospective log.
V02 Live scoring/postponements	PARTIAL (tests only)	Synthetic scoring/unscored separation tested; real log/result join and postponed fixtures not verified.
R01 Fresh reproducible setup	PARTIAL	Source versions, commands, saved run and 38-test log exist; fresh clean installation not repeated in this report audit.
R02 Documents/figures/slides	PARTIAL / OPEN	Current numbers agree; stale missing-document text and matrix title mismatch disclosed; no personal approval.
R03 Course compliance	PENDING	Part 2 headings reconciled; names, dates, signed artifacts, topic uniqueness and approvals still need group evidence.

Release decision remains pending personal review. In particular, D06, M02, the unverified planned edge cases in M03/P01/E01, real live evidence in V01/V02, fresh-clean setup in R01, and manual consistency/compliance in R02/R03 cannot be promoted to full passes merely because 38 tests passed. No new tests were added and no existing output was overwritten. The independent comparison is **partial**: all 20 ranks, W/D/L records and points agree with the preserved NBC transcription, and 19 goal differences agree. Fulham is printed as +4 there versus -4 from 47 GF and 51 GA. The source discrepancy is preserved. Official P/GF/GA and administrative adjustments are still unverified; the report does not infer that the publication has been corrected.

Sources: qa/Test_Plan.md, outputs/test-results.txt, qa_checks.json, source_metadata.json, stored prediction/forecast CSVs, qa/Ankit_Karki_QA_Workbook.xlsx, qa/Independent_Table_Comparison.csv and the unchanged test source.

Quality Assurance Signoff

6.2 Error Analysis

The following ledger reproduces all source findings and fix/actions, adding a causal explanation only where supported. Unknown connection/publication causes are bracketed rather than invented. All entries are dated October 2, 2026. Historical finding text is preserved even when its old assessment was wrong. **QA-016 confirms the Results Requirements are in the repository and resolves QA-007's availability claim.** QA-017 records the newly found matrix-title mismatch; code and outputs are frozen as requested.

ID / severity · finding and cause	Fix or action · evidence	Current status
QA-001 · High Finding: Pasted draft uses Pinnacle closing odds despite provider staleness warning Cause: Pinnacle API staleness made the pasted benchmark unsuitable.	Use fixed AvgCH AvgCD AvgCA triplet and valid common sample Evidence: https://www.football-data.co.uk/data.php	Verified fixed AvgC runtime benchmark
QA-002 · High Finding: Fixed 0-10 goal grid can omit material mass at high rates Cause: Finite fixed cap does not bound the omitted tail for every fitted rate.	Expand grid until omitted mass below 1e-8 and record tail Evidence: Proposal mathematical approach	Verified adaptive tail tests and actual run
QA-003 · High Finding: First-190-row season cutoff can split a date Cause: Row-count splitting ignores simultaneous date batches.	Use first complete date boundary at or after 190 results Evidence: Proposal season backtest	Verified Jan 1 complete-date boundary
QA-004 · Medium Finding: Claiming exactly 280 predictions can conflict with date-only warmup handling Cause: A nominal warm-up index was confused with a full-date evaluation boundary.	Skip full warmup date and report actual evaluated count Evidence: Proposal walk-forward protocol	Verified 280 eligible rows for this snapshot
QA-005 · Medium Finding: Three-SE simulation test and 90 percent interval coverage were phrased as guaranteed passes Cause: Random diagnostics were interpreted as deterministic acceptance promises.	Treat random discrepancy and empirical coverage as diagnostics not guaranteed outcomes Evidence: qa/Test_Plan.md	Corrected in QA plan
QA-006 · High Finding: Other member names and Canvas due dates were not supplied Cause: Required group/scheduling information was not supplied.	Leave clearly marked fields for group to complete Evidence: docs/Proposal.md	Pending group input
QA-007 · High Finding: Part 2 Results Requirements are absent from repository Cause: Earlier inventory omitted an existing root PDF; current file is confirmed, including in original baseline history.	Read root Results Requirments.pdf and reconcile Part 2 report; availability confirmed by QA-016 Evidence: Results Requirments.pdf; QA-016; deliverables/Results_Report.pdf	Resolved availability/inventory error; see QA-016; human completion remains pending
QA-008 · Medium Finding: Rebuilding the same CSV is not an independent official table verification Cause: Reusing one CSV is circular, not an independent table check.	Obtain authoritative 2025/26 table and reconcile numbers Evidence: QA case D06	Partial independent comparison; official full table still pending
QA-009 · Low Finding: Direct curl TLS retrieval failed in this sandbox Cause: TLS retrieval failed; underlying connection cause [not recorded].	Downloaded historical CSV via browser; verified equality with extraction copy Evidence: data/provenance.json	Resolved for historical snapshot
QA-010 · Medium Finding: Live CSV snapshot includes only results through Sep 20 and no saved prospective forecast Cause: Past result snapshot has no immutable, advance-saved forecast evidence.	Keep extension optional and do not call past results a live forecast Evidence: data/raw/E0_2627.csv	Open extension dependency

Source ledger: qa/Issue_Log.csv. "Resolved" for a technical defect does not imply course approval or a student's signature.

Quality Assurance Signoff — Error Analysis continued

ID / severity · finding and cause	Fix or action · evidence	Current status
QA-011 · Medium Finding: Public repo and provider reuse conditions require raw-data handling Cause: Public storage must distinguish code/derived work from provider raw data.	Ignore raw CSVs in public Git; use private coursework copies and attribution Evidence: https://www.football-data.co.uk/data.php	Verified raw CSVs ignored before public branch push
QA-012 · Low Finding: Pasted claim of no xG does not hold for the inspected 2026/27 schema Cause: Live-season schema differs from the pasted assumption.	HxG and AxG are present but intentionally excluded from both core models Evidence: docs/Proposal.md	Documented scope decision
QA-013 · Medium Finding: Independent NBC article states Fulham GD as positive 4 but frozen scores give negative 4 Cause: Publication sign differs from GF-GA; publication cause [unverified].	Preserve discrepancy; all 20 ranks W D L points agree and other 19 GD agree; verify official full table Evidence: qa/Independent_Table_Comparison.csv	Pending authoritative resolution
QA-014 · Medium Finding: LibreOffice recalculation of unprefix POISSON.DIST produced NAME errors Cause: Unprefixed function name was not accepted in LibreOffice recalculation.	Replace workbook grid formulas with direct EXP powers FACT Poisson definition and recalculate Evidence: scripts/build_qa_workbook.py	Resolved; zero formula errors and all five rate checks agree
QA-015 · Medium Finding: Coding worker unexpectedly pushed partial live refinement to main instead of leaving local changes Cause: Worker did not respect the local-only integration boundary.	Restored original main contents with regular revert 6d621c7; complete project isolated on review branch Evidence: Git main repair commit 6d621c7	Resolved; verify original main diff empty
QA-016 · High Finding: Results Requirements document is now confirmed in the repository; earlier missing-document claim was incorrect Cause: The former missing-document assessment required correction and rubric reconciliation.	Read Results Requirments.pdf and align the Results report to every required heading; resolves QA-007 availability Evidence: Results Requirments.pdf; deliverables/Results_Report.pdf	Resolved document availability; approval and personal review not implied
QA-017 · Medium Finding: Selected-score-matrix PNG title says shown 0-10 but the saved matrix is 16 by 16 with goal indices 0-15 Cause: Plot uses the full adaptive matrix but retains a fixed-range title/tick selection.	Preserve source PNG/code/outputs unchanged; disclose exact extent in report caption; later crop to 0-10 or label 0-15 and regenerate only with authorization Evidence: outputs/selected_score_matrix.csv; outputs/figures/selected_score_matrix.png; src/standings/reporting.py	Open figure-label correction; disclosed in Results report

How the open items will be closed. The Lead supplies names/dates and confirms topic uniqueness, contract, peer forms and approval (QA-006/R03). Ankit obtains a full authoritative table, checks P/W/D/L/GF/GA/GD/Pts, and resolves the retained sign discrepancy (QA-008/013). Live work stays optional unless actual future fixtures and immutable advance logs are obtained (QA-010); Model 2 and shrinkage tuning remain uncompleted, not silently promised. After authorization, the Planner fixes the matrix extent label and regenerates only separately preserved evidence (QA-017). Planned edge checks, fresh setup, personal spreadsheet checks, slide export and rehearsal then need recorded reviewers and dates. The old proposal/results/checklist missing-document statements are stale; this report corrects them without rewriting the original artifacts.

QA Lead: Ankit Karki — personal sign-off only after review.

Signature: _____

Date: _____

References [1] Maher, M. J. (1982). “Modelling association football scores.” *Statistica Neerlandica*, 36(3), 109–118. DOI: [10.1111/j.1467-9574.1982.tb00782.x](https://doi.org/10.1111/j.1467-9574.1982.tb00782.x).

[2] Dixon, M. J., and Coles, S. G. (1997). “Modelling association football scores and inefficiencies in the football betting market.” *JRSS C*, 46(2), 265–280. [University citation](#). Model 2 is not implemented.

[3] [football-data.co.uk](https://www.football-data.co.uk/). [Data description/conditions](#), [England files](#), [field notes](#). Frozen local evidence as supplied.

[4] SciPy documentation: [scipy.stats.poisson](#) and [scipy.stats.skellam](#).